

Compiler Construction (CS-4031)

Sessional-II Exam

Date: 04/14/2026

Total Time: 1 hr

Course Instructors

Total Marks: 30

Shaharbano Asif, Muhammad Kashif, Zulfiqar Ali, Ghulam Murtaza

Total Questions: 03

Roll No

Section

Student Signature

Do not write below this line.

Attempt all questions.

CLO # 1: Understand the basic techniques used in compiler construction such as lexical analysis, top-down, bottom-up parsing, context-sensitive analysis, and intermediate code generation.

Question 1: Provide brief answers to the following:

(5 x 2 = 10 points)

- a. What is the key difference between Top-Down and Bottom-Up parser?

Answer: Top-down parsers build the parse tree from the start symbol to the input (using leftmost derivation), while bottom-up parsers reduce the input to the start symbol (using rightmost derivation in reverse).

- b. What does it mean for a grammar to be ambiguous? Give one ambiguous grammar.

Answer: A grammar is ambiguous if there exists a string with two or more distinct parse trees (or leftmost derivations). Example: $S \rightarrow S + S \mid S * S \mid id$.

- c. Briefly Discuss shift-reduce conflict?

Answer: A shift-reduce conflict occurs when a parser cannot decide whether to shift (push a terminal) or reduce (apply a production) based on the next input symbol and stack contents.

- d. Where and why do we need left factoring?

Answer: Left factoring is needed to eliminate common prefixes in productions for the same nonterminal, making the grammar suitable for predictive top-down parsing by avoiding backtracking.

- e. Which is stronger among CLR and LALR? Why?

Answer: CLR (LR(1)) is stronger than LALR because it uses full lookahead sets, while LALR merges states with identical cores, which may introduce reduce reduce conflicts but makes tables smaller.

Question 2:

(5 + 5 = 10 points)

For the given Grammar

$$A \rightarrow BD$$

$$B \rightarrow twB \mid tuwB \mid \epsilon$$

$$D \rightarrow Dx B \mid v$$

- a) Examine the grammar and rewrite it so that an LL(1) predictive parser can be built for the corresponding language.

Q#02 (a)

$$A \rightarrow BD$$

$$B \rightarrow twB \mid tuwB \mid \epsilon$$

$$D \rightarrow Dx B \mid v$$

Left Factoring

$$A \rightarrow BB$$

$$B \rightarrow tB' \mid \epsilon$$

$$B' \rightarrow wB \mid uwB$$

$$D \rightarrow Dx B \mid v$$

Remove Left Recursion:

$$D \rightarrow Dx B \mid v$$

$$D \rightarrow vD'$$

$$D' \rightarrow xBD' \mid \epsilon$$

Final Grammar

$$A \rightarrow BD$$

$$B \rightarrow tB' \mid \epsilon$$

$$B' \rightarrow wB \mid uwB$$

$$D \rightarrow vD'$$

$$D' \rightarrow xBD' \mid \epsilon$$

- b) Compute the FIRST and FOLLOW sets for all non-terminal symbols in the new grammar and build the parse table.

First

$$A \rightarrow t \mid v$$

$$B \rightarrow \epsilon$$

$$B' \rightarrow w \mid u$$

$$D \rightarrow v$$

$$D' \rightarrow x \mid \epsilon$$

FOLLOW

$$A \rightarrow \{ \$ \}$$

$$B \rightarrow \{ v, x, \$ \}$$

$$B' \rightarrow \{ v, x, \$ \}$$

$$D \rightarrow \{ \$ \}$$

$$D' \rightarrow \{ \$ \}$$

Q# 2(B) LR(0) for LR

	ϵ	w	u	v	x	\$
A	$A \rightarrow \epsilon D$ $A \rightarrow B$			$A \rightarrow BD$		
B	$B \rightarrow \epsilon B'$			$B \rightarrow \epsilon$	$B \rightarrow \epsilon$	$B \rightarrow \epsilon$
B'		$B' \rightarrow wB$	$B' \rightarrow u w B$			
D				$D \rightarrow v D'$		
D'					$D' \rightarrow x B D'$	$D' \rightarrow \epsilon$

CLO # 3: Design and implement a compiler using a software engineering approach.

Question 3:

(5 + 5 = 10 points)

For the given Grammar

$\langle \text{DeclList} \rangle \rightarrow \langle \text{Decl} \rangle$

$\langle \text{Decl} \rangle \rightarrow \langle \text{Type} \rangle \text{ id};$

$\langle \text{type} \rangle \rightarrow \text{int} \mid \text{float}$

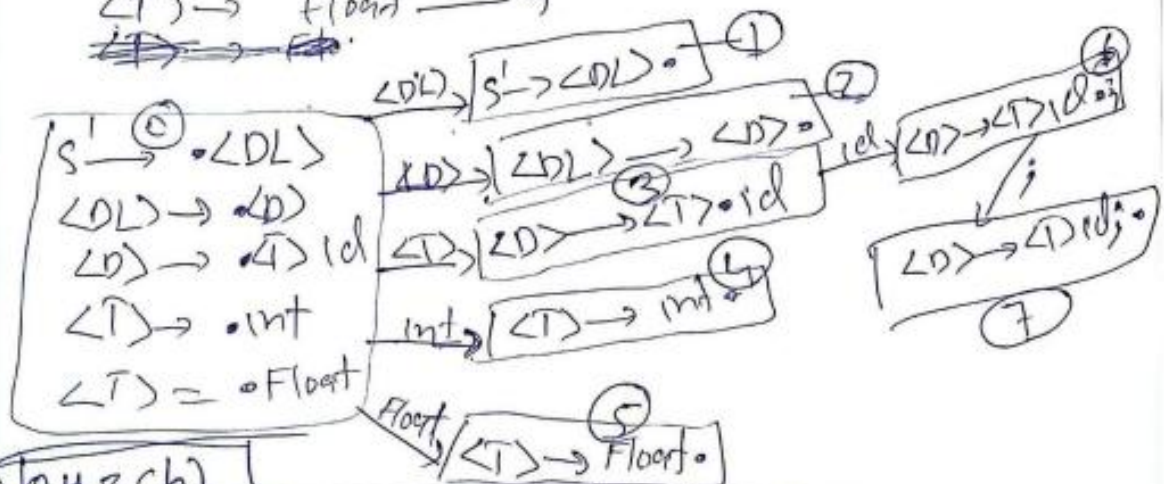
- Obtain the canonical items.
- Find the LR-Parsing table and justify either it is accepted by LR(0) or not.

$\langle \text{DeclList} \rangle \rightarrow \langle \text{Decl} \rangle \rightarrow \langle \text{DeclList} \rangle \rightarrow \langle \text{DL} \rangle$
 $\langle \text{Decl} \rangle \rightarrow \langle \text{type} \rangle \text{id} \rightarrow \langle \text{Decl} \rangle \rightarrow \langle \text{D} \rangle$
 $\langle \text{type} \rangle \rightarrow \text{int} / \text{float} \rightarrow \langle \text{type} \rangle \rightarrow \langle \text{T} \rangle$

Q3(a)

Augmented Grammar

$S' \rightarrow \cdot \langle \text{DL} \rangle \rightarrow 0$
 $\langle \text{DL} \rangle \rightarrow \cdot \langle \text{D} \rangle \rightarrow 1$
 $\langle \text{D} \rangle \rightarrow \cdot \langle \text{T} \rangle \text{id}; \rightarrow 2$
 $\langle \text{T} \rangle \rightarrow \text{int} \mid \text{float} \rightarrow 3$
 $\langle \text{T} \rangle \rightarrow \text{float} \rightarrow 4$



Q3(b)

States	Actions				Goto			Note
	id	int	float		$\langle \text{DL} \rangle$	$\langle \text{D} \rangle$	$\langle \text{T} \rangle$	
0				S5	1	2	3	Note: This error is acceptable by LRO
1								
2	R1	R1	R1	R1				
3	S6							
4	R3	R3	R3	R3				
5	R4	R4	R4	R4				
6	R2	R2	R2	R2				
7	R2	R2	R2	R2				